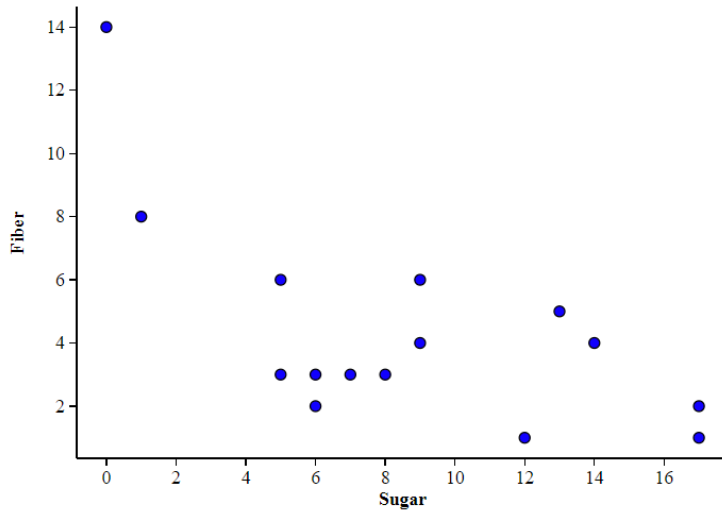


1) Fiber helps regulate the body's use of sugars, helping to keep hunger and blood sugar in check. In children's cereal, which is usually loaded with large amounts of sugar, scientists wanted to investigate if a larger sugar content led to less nutritional value overall, such as lower fiber. A scatter plot of 15 randomly selected children's cereals, along with some selected summary statistics, are given below. Both sugar and fiber are measured in grams per serving.



Sugar	Fiber
$\bar{x} = 8.6$	$\bar{y} = 4.33$
$S_x = 5.18$	$S_y = 3.31$
$r = -0.654$	

- a) Does the scatterplot indicate that it is okay to create a linear model? Explain.
- b) Find the slope of the least-squares regression line. Interpret this value in context.
- c) What point must be on the least-squares regression line?
- d) Find the intercept of the least-squares regression line. Interpret this value in context.

e) Write the equation of the linear model.

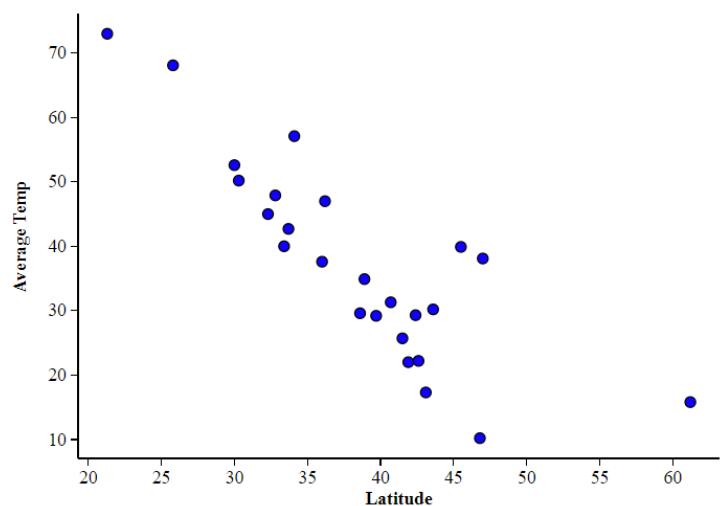
f) If you pick up a cereal and see that it has 3 grams of sugar, what is the predicted fiber content?

g) What is the residual for the cereal with 9 grams of sugar and 6 grams of fiber?

h) Calculate the value of r^2 and interpret this value in context.

2) We know that the further you get from the equator, the colder the climate becomes. For the follow cities, we have assembled their latitude (angular distance from the equator, measured in degrees) and their average temperature in December of 2021. The data and the scatterplot are given below.

City	Latitude (°N)	Dec Average Temp (°F)
Albany, NY	42.6	22.2
Anchorage, AK	61.2	15.8
Atlanta, GA	33.7	42.7
Austin, TX	30.3	50.2
Bismarck, ND	46.8	10.2
Boise, ID	43.6	30.2
Boston, MA	42.4	29.3
Charleston, SC	32.8	47.9
Chicago, IL	41.9	22
Cleveland, OH	41.5	25.7
Denver, CO	39.7	29.2
Honolulu, HI	21.3	73
Jackson, MS	32.3	45
Knoxville, TN	36	37.6
Las Vegas, NV	36.2	47



Los Angeles, CA	34.1	57.1
Madison, WI	43.1	17.3
Miami, FL	25.8	68.1
Newark, NJ	40.7	31.3
New Orleans, LA	30	52.6
Olympia, WA	47	38.1
Portland, OR	45.5	39.9
Roswell, NM	33.4	40
St. Louis, MO	38.6	29.6
Washington, DC	38.9	34.9

a) Describe the relationship between latitude and average December temperature in the US.

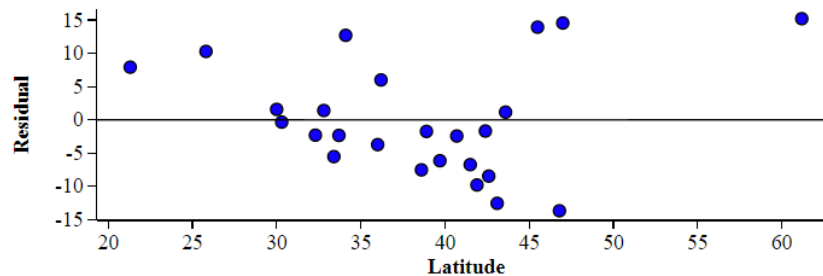
b) Find the LSRL using your calculator.

c) Find the following summary statistics using your calculator:

$$\bar{x} = \text{_____} \quad \bar{y} = \text{_____} \quad S_x = \text{_____} \quad S_y = \text{_____} \quad r = \text{_____}$$

d) Using the summary statistics from (c), show that the LSRL you find by hand matches what you found with your calculator in (b).

e) Below is the residual plot for latitude and average December temperature.



Given the residual plot above, is a linear regression model appropriate for this data set? Explain.

f) Kansas City, MO is located at 39.1°N and had an average December 2021 temperature of 42°F. What is the residual for this point?